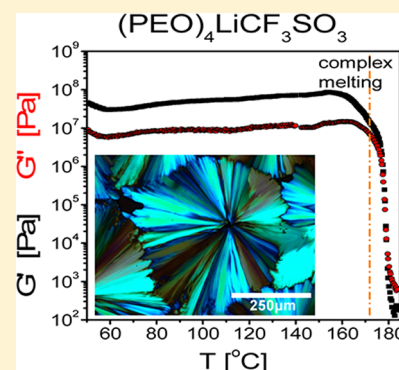


Relating Structure, Viscoelasticity, and Local Mobility to Conductivity in PEO/LiTf Electrolytes

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S Supporting Information

ABSTRACT: The phase state, local structure, local mobility, and viscoelastic response have been studied in the archetypal polymer electrolyte (PEO)_xLiCF₃SO₃ with ether oxygen to lithium ion ratio of $2 \leq [\text{EO}]/[\text{Li}] \leq 12$ over a broad temperature range in an effort to explore the factors controlling ionic conduction. We confirm that the crystal structure of the complex is identical to the (PEO)₃LiCF₃SO₃ polymer electrolyte independent of the [EO]:[Li] content. Heating the nonstoichiometric compositions result in progressive melting of the complex, whereas the complex formed at or near the stoichiometric composition remains stable up to the liquidus temperature. The temperature dependence of dc conductivity is neither Arrhenius nor VFT. Its temperature dependence is more complex reflecting the underlying structural changes. Surprisingly, ionic conduction takes place both within the crystalline complex and in the amorphous phase with the latter having the major contribution. The (PEO)₁₂LiCF₃SO₃ polymer electrolyte is the one with the highest conductivity at all temperatures investigated. The linear viscoelastic properties were studied as a function of temperature at two compositions. The different phases have distinct viscoelastic signatures. The complex formed at or near stoichiometric composition has a predominantly *elastic* response, whereas the more dilute compositions (consisting of the crystalline complex and an ion-containing amorphous phase) have a *viscoelastic* response and an ultraslow relaxation. Local polymer relaxation and ionic mobility are completely coupled. It is suggested that local ion jumps at subsegmental level are responsible for the measured conductivity.



I. INTRODUCTION

Liquid-based lithium (Li) batteries dominate the current market.^{1–3} This is based on the fact that Li is the lightest of all metals, and when used as an anode in contact with Li⁺–salt electrolytes, it provides a wider electropositive potential window. However, concerns about safety hazards (solvent leakage, flammability) as well as limited temperature range of operation and electrode corrosion have fostered research in all-solid-state rechargeable Li batteries. Solid-state batteries offer unique opportunities for greatly increased cycle life, safety, and energy density. Polymeric electrolyte materials have been investigated for the past 25 years as promising materials for electrochemical device applications including high energy density rechargeable batteries, fuel cells, supercapacitors, electrochromic displays, etc. The first ion conducting polymer was based on a poly(ethylene oxide) (PEO)–Li–salt complexed solid polymer electrolyte (SPE) membrane.⁴ Since then, different alternatives have been investigated that include gels, plasticized polymers, and composite polymer electrolytes. A reliable device requires polymer electrolytes that combine high ionic conductivity at ambient temperature ($\sigma > 10^{-4}$ S/cm), high ionic transference number (preferably cationic), mechanical stability, chemical, thermal, and electrochemical stability, and compatibility with the electrode materials. Although major research has been conducted on SPEs,^{5–8} the

phenomenon of ion transport is not completely understood mainly because charge transport is intimately connected to a number of structural features encountered in these systems.

Among the different SPEs, the *archetypal* polymer electrolyte with potential applications is LiCF₃SO₃ dissolved in PEO.^{9–21} This reflects on the ability of PEO to effectively dissolve and complex the ionic salt. Among all compositions the one more extensively studied, and referred to as the stoichiometric, is the (PEO)₃LiCF₃SO₃ with an ether oxygen to lithium ion ratio [EO]:[Li⁺] = 3:1.¹⁰ At this composition the PEO chains adopt a helical conformation with Li⁺ ions located in each turn of the helix and coordinated by three ether oxygens. Furthermore, each Li⁺ ion coordinates with an oxygen from two CF₃SO₃ anions, where each CF₃SO₃ anion bridges two adjacent Li⁺ ions along the chain. These ionic chains are packed regularly to form a monoclinic unit cell. In addition to these structural features, the CF₃SO₃ (triflate, Tf) anion has intramolecular vibrational modes that are sensitive to ionic association, making it possible to determine the various ionic species by IR spectroscopy.^{16–18}

The phase diagram of the (PEO)_xLiCF₃SO₃ polymer electrolyte is shown in Figure 1 adapted from ref 9. At

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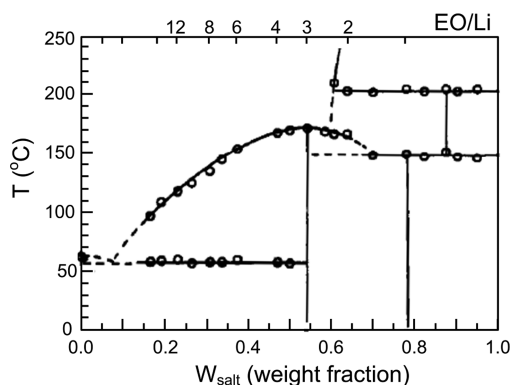


Figure 1. Phase diagram for $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolyte system ($M_w(\text{PEO}) = 3900$ g/mol) reproduced from ref 9.

temperatures below 60 °C (solidus) it consists of crystalline PEO, crystalline $(\text{PEO})_3\text{LiCF}_3\text{SO}_3$ complex, and possibly an “amorphous” phase whose characteristics are debated. At temperatures above 60 °C, PEO melts and the complex starts to dissolve gradually to the amorphous phase (sometimes called “elastomeric phase”) until it disappears completely at the liquidus temperature (in the range from 100 to 180 °C depending on composition). At higher temperatures the polymer electrolyte forms a single amorphous phase.

The amorphous phase plays a key role in charge transport; hence, there have been efforts to determine the kind of charged species that contribute to conduction. IR and Raman spectroscopies^{16–18,22,23} identified free triflate anions, free Li^+ ions solvated by PEO, contact ion pairs, and aggregates $[\text{Li}_2\text{CF}_3\text{SO}_3]^+$ that are appointed to the $(\text{PEO})_3\text{LiCF}_3\text{SO}_3$ compound as the main charged entities. At high temperatures and low salt concentrations the prevailing ionic species are the free triflate anions and the corresponding Li^+ solvated ions. Furthermore, IR revealed a decreasing number of free triflate anions and a dominate role of contact ion pairs and aggregates with increasing concentration.^{16,22,23} At and above the liquidus temperature both triflate anions and Li^+ are mobile, but the former are only weakly interacting with the polymer in contrast to the strong interaction of Li^+ .¹² In fact, it has been suggested that the Li^+ –polyether backbone is not significantly altered upon melting, indicating a strong interaction even in the amorphous phase.¹⁶ In view of the different interactions, the transport properties for the two charge carriers are expected to be significantly different. This was supported by pulsed-field-gradient NMR measurements¹² of the cation and anion diffusion coefficients in the same $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolyte.

Despite these important studies, there have been only limited efforts to connect the measured conductivities to the underlying structural and thermodynamic properties in the archetypal $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolyte. Surprisingly, the viscoelastic properties of the complex and of the amorphous phase that can influence the device properties have not been explored. In addition, the role of the crystalline complex in ionic conduction and the possible ionic mechanism remain largely unexplored. In the present study we employ differential scanning calorimetry, X-ray scattering, dielectric spectroscopy, and rheology and investigate the crystal structure at the different compositions, the viscoelastic response of the different phases (crystalline complex and amorphous phase), and the dynamics give rise to the ionic conductivity.

II. EXPERIMENTAL SECTION

Samples. Synthesis of Poly(ethylene oxide) Homopolymers.

Poly(ethylene oxide) homopolymers were synthesized by anionic polymerization high-vacuum techniques using methodologies described elsewhere.^{24,25} Briefly, *sec*-BuLi was used as the initiator for the polymerization of purified ethylene oxide (distilled over CaH_2 , and twice over *n*-BuLi) in the presence of phosphazene base (used as received). After addition of phosphazene solution in benzene the temperature was raised to 40 °C and allowed for 48 h for completion of the ethylene oxide polymerization. The reaction was terminated by adding degassed methanol to the polymerization solution, followed by a drop of HCl. The homopolymers were isolated by precipitation into a large excess of hexane and dried under vacuum for several days. The molecular characteristics of the PEO homopolymers utilized are summarized in Table 1.

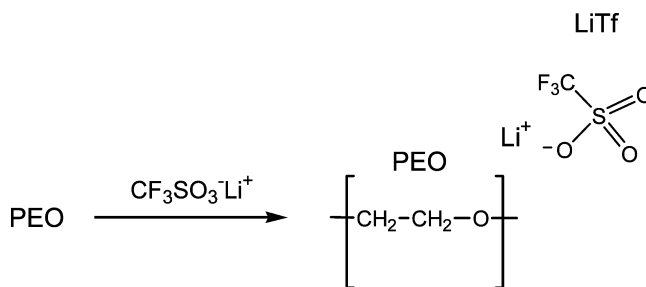
Table 1. Molecular Characteristics of Poly(ethylene oxide) Homopolymers Utilized in This Study

sample	M_w^a (g/mol)	M_w/M_n^a
PEO14	14 000	1.06
PEO22	22 000	1.05
PEO50	50 000	1.10

^aDetermined by SEC in THF.

Preparation of PEO:LiTf Mixtures. At first, 5% w/v solutions of PEO homopolymers of different molecular weights were prepared in tetrahydrofuran, and each one was separated to six vials. Also, six 5% w/v solutions of different weighed amounts of LiCF_3SO_3 (LiTf) were prepared in tetrahydrofuran. Each homopolymer solution was mixed with the corresponding salt solution to form $[\text{EO}]:[\text{Li}]$ doping ratios of 12:1, 8:1, 6:1, 4:1, 3:1, and 2:1. The resulting homopolymer electrolyte mixtures were isolated by evaporation of the solvent and dried to a constant weight under vacuum. The preparation route for salt incorporation is presented in Scheme 1.

Scheme 1



Differential Scanning Calorimetry (DSC). The thermal behavior of the polymer electrolytes was studied by differential scanning calorimetry on cooling and subsequent heating at a rate of 10 K/min with a Mettler 30 DSC. The DSC traces from the second cooling and heating scans of each polymer electrolyte with different molecular weights are shown in Figures 2 and 3 and in the Supporting Information (Figure S1).

Thermogravimetric Analysis (TGA). The thermal stability of the polymer electrolytes with compositions 4:1 and 8:1 and PEO molecular weight of 50 kg/mol were studied with a TGA, and the results are depicted in Figure S2. The result shows a stable weight up to 280 °C.

Polarizing Optical Microscopy (POM). A Zeiss Axioskop 40 equipped with a video camera and a fast frame grabber was used to follow the superstructure formation in two mixtures $(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ and $(\text{PEO})_{12}\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol). A Linkham temperature control unit (THMS600) equipped with a TMS94 temperature programmer was employed for the temperature-dependent studies.

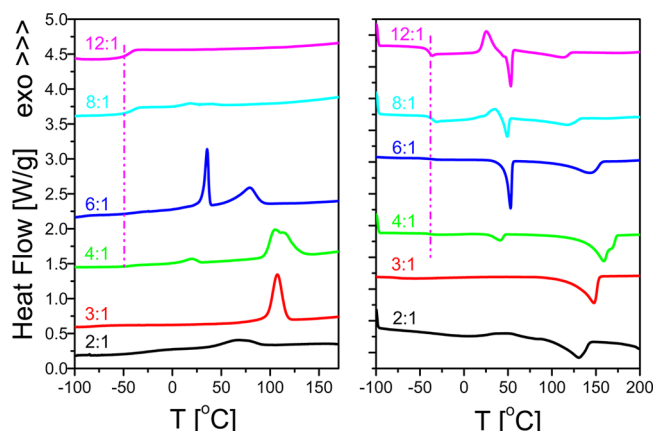


Figure 2. DSC traces of the polymer electrolytes $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ ($M_w = 14$ kg/mol) with x ranging from 2:1 to 12:1, on cooling (left) and subsequent heating (right) with a rate of 10 K/min. The vertical lines indicate the approximate position of the glass temperature (see text).

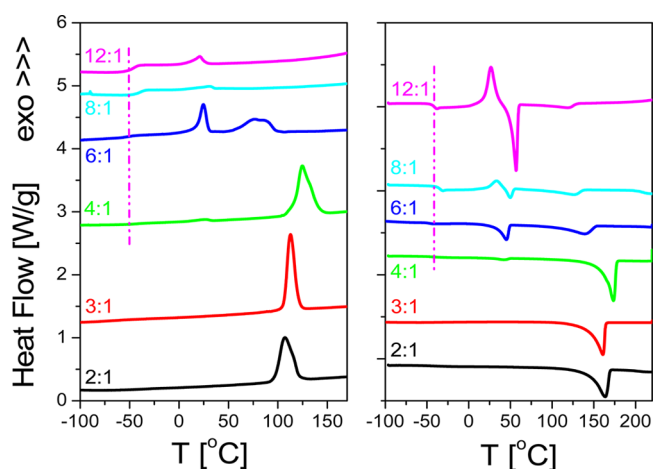


Figure 3. DSC traces of the polymer electrolytes $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol) with x ranging from 2:1 to 12:1, on cooling (left) and subsequent heating (right) with a rate 10 K/min. The vertical lines indicate the approximate position of the glass temperature (see text).

Images were recorded following slow cooling (1 K/min) from the melt state.

X-ray Scattering. Wide-angle X-ray scattering (WAXS) measurements have been performed on powder samples. The $\Theta/2\Theta$ scans were taken with a D8 Advance X-ray diffractometer (Bruker). The X-ray tube (Kristalloflex 780) generator with a Cu anode was operating at a voltage of 40 kV and a current of 30 mA. An aperture (divergence) slit of 0.3 mm and a scattered-radiation (antiscatter) slit of 0.3 mm together with a monochromator slit of 0.1 mm and a detector slit of 1 mm were employed. A diffracted beam monochromator was inserted between the detector slit and the detector to suppress fluorescence radiation and the unwanted $K\beta$ radiation. The monochromator employed a graphite crystal ($2d^* = 0.6714$ nm, for the 002 reflection). The $K\alpha_1$ and $K\alpha_2$ peaks could not be separated, and an average wavelength of 0.154 184 nm was used. A scintillation counter with a 95% quantum yield for the Cu radiation was employed as the detector. Scans in the 2Θ range from 1° to 40° in steps of 0.01° were made under vacuum at different temperatures ranging from 298 to 453 K depending on the sample.

Rheology. Two rheometers were used for recording the viscoelastic properties of two polyelectrolytes. For $(\text{PEO})_{12}\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol) an Advanced Rheometric Expansion System (ARES) equipped with a force-rebalanced transducer was used in the

oscillatory mode. Depending on the temperature range, two transducers were used with 2000, 2 g·cm and 200, 0.2 g·cm upper and lower sensitivity, respectively. For the $(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol) polymer electrolyte, a TA Instruments (model AR-G2) with a magnetic bearing that allows for low nanotorque control was employed. Measurements were made with the environmental test chamber as a function of temperature. In both cases, samples were prepared on the lower plate of the 8 mm diameter parallel plate geometry setup and heated under a nitrogen atmosphere until it could flow. Subsequently, the upper plate was brought into contact, the gap thickness was adjusted to 1 mm, and the sample was slowly cooled to the desired starting temperature. The storage (G') and loss (G'') shear moduli were monitored in different types of experiments. First, the linear and nonlinear viscoelastic ranges were identified by recording the strain amplitude dependence of the complex shear modulus $|G^*|$ at selected temperatures (at $\omega = 10$ rad/s). In subsequent experiments strain amplitudes within the linear viscoelastic range were used. These experiments involved isothermal frequency scans within the range $10^{-1} < \omega < 10^2$ rad/s at some selected temperatures that correspond to the crystalline PEO, crystalline complex, and melt states.

Dielectric Spectroscopy (DS). All sample capacitors were prepared in a glovebox of controlled N_2 atmosphere (O_2 and H_2O content of less than 0.1 ppm) to avoid water intake and oxidation during preparation. The samples were heated within the glovebox just below the melting temperature of the complex, deposited on gold-plated electrodes, and slightly pressed in order to uniformly spread between the electrodes and optimize the contact. Measurements were made for different sample thicknesses (thickness of the capacitors was maintained by Teflon spacers), but the results reported in the main text were obtained with a 1 mm spacer. The effect of sample thickness on the slow mechanism associated with the buildup of charges at the electrode surface (known as electrode polarization, EP) is examined in Figure S3. Dielectric measurements were made with a Novocontrol Alpha frequency analyzer under “isobaric” conditions as a function of temperature. Measurements were performed at different temperatures in the range 213.15–453.15 K in steps of 5 K, at atmospheric pressure, and for frequencies in the range from 10^{-2} to 10^7 Hz. The complex dielectric permittivity $\epsilon^* = \epsilon' - i\epsilon''$, where ϵ' is the real and ϵ'' is the imaginary part, has been obtained as a function of frequency ω and temperature T , i.e., $\epsilon^*(T, \omega)$.^{26,27} The analysis of the T -dependent experiments was made using the empirical equation of Havriliak and Negami (HN):²⁸

$$\epsilon_{\text{HN}}^*(\omega, T) = \epsilon_\infty(T) + \frac{\Delta\epsilon(T)}{[1 + (i\omega\tau_{\text{HN}}(T))^m]^n} + \frac{\sigma_0(T)}{i\epsilon_f\omega} \quad (1)$$

where $\epsilon_\infty(T)$ is the high-frequency permittivity, $\tau_{\text{HN}}(T)$ is the characteristic relaxation time in this equation, $\Delta\epsilon(T) = \epsilon_0(T) - \epsilon_\infty(T)$ is the relaxation strength, m and n (with limits $0 < m, mn \leq 1$) describe respectively the symmetrical and asymmetrical broadening of the distribution of relaxation times, σ_0 is the dc conductivity, and ϵ_f is the permittivity of free space. From τ_{HN} , the relaxation time at maximum loss, τ_{max} , is obtained analytically following

$$\tau_{\text{max}} = \tau_{\text{HN}} \left[\frac{\sin\left(\frac{\pi m}{2 + 2n}\right)}{\sin\left(\frac{\pi mn}{2 + 2n}\right)} \right]^{-1/m} \quad (2)$$

The measured ϵ'' spectra and the derivative of ϵ' ($d\epsilon'/d \ln \omega \sim -(2/\pi)\epsilon''$) have been used for the analysis of the dynamic behavior with respect to the segmental relaxation. Conductivity studies were made using the analysis of the complex conductivity function $\sigma^* = \sigma' + i\sigma''$, which is related to the complex dielectric permittivity with $\sigma^* = i\epsilon\omega_0\epsilon^*$. To obtain the dc conductivity, two methods were employed. First, we employed the random free energy barrier model by Dyre.²⁹ The model assumes that conduction takes place by hopping, where the hopping charge carriers are subject to spatially randomly varying energy barriers. According to the model, the onset of dc conductivity is determined by crossing the highest barrier. The model, solved in the continuous time random walk approximation, provides an analytical

expression for the complex dielectric function as^{29,30} $\epsilon^*(\omega) = \epsilon_\infty + \sigma_0 \tau_e / [\epsilon_0 \ln(1 + i\omega\tau_e)]$, where ϵ_∞ is the value of ϵ' in the limit of high frequencies and σ_0 , τ_e are the dc conductivity and characteristic time of ion motion. On the basis of only these two parameters, it predicts a universal shape for the conductivity contribution. The model predictions are tested against the experimental data at three temperatures for the $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ compound in Figure S4. The predictions can only partially fit the experimental data and within a limited frequency range. We attribute these deviations to the nature of polymer electrolyte; the system cannot be considered as disordered, and in addition, the complex is strongly temperature-dependent. Nevertheless, the thus extracted characteristic relaxation times, τ_e , scale as $\sigma_0 \sim \tau_e^{-1}$ in accord with the prediction of the Barton–Nakajima–Namikawa (BNN) relation.²⁶ This is depicted in Figure S5. Second, the dc conductivity can be obtained by the plateau in the real part, σ' without invoking any model (see Figure S6). In this case, the rates of ionic motion (ion mobility) were determined by the analysis of the complex electric modulus $M^* = M' + iM''$, which is related to the complex dielectric permittivity as $M^* = 1/\epsilon^*$. In particular, in samples with strong conductivity, the characteristic time of ion motion was extracted from the position of the maximum value in M'' . Alternatively, the characteristic time can be obtained from the crossing of the real and imaginary parts. The two methods give identical relaxation times for narrow (Debye) processes but differ for broader processes.²⁷ The effect of thermal treatment on the thus extracted dc conductivity is examined in Figure S7.

III. RESULTS AND DISCUSSION

Thermodynamics and Phase State. The phase diagram of $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolyte (with PEO molecular weight of 3900 g/mol) is depicted in Figure 1. The DSC traces of $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ with molecular weights of 14, 22, and 50 kg/mol are discussed with respect to Figures 2, 3, and S1. All traces refer to the second cooling and heating runs with a rate of 10 K/min. For the 14 kg/mol $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ (Figure 2), the cooling trace of the 12:1 compound shows only a glass temperature at -47°C . An exothermic peak due to complex formation is not clearly seen. On heating, the glass temperature appears at -42°C and is followed by an exothermic peak reflecting the PEO cold crystallization/complex formation at 25°C followed by PEO melting endotherm at 53°C . At higher temperatures a broad and very asymmetric endothermic process exists with a peak at 113°C , signifying the melting of the complex. The cooling and heating traces of the 8:1 compound are qualitatively similar, but now the complex melts at a somewhat higher temperature (118°C). The cooling trace of the 6:1 compound shows a broad exothermic peak due to complex formation with a peak at 79°C and a PEO crystallization peak at 35°C followed by a weak liquid-to-glass step at about -38°C . On heating, the trace shows the PEO melting endotherm at 53°C followed by a very broad endothermic process signifying the continuous melting of the complex with a peak at 144°C . The traces of the compounds with ratios 4:1, 3:1, and 2:1 are qualitative similar in that they reveal that PEO is unable to crystallize apart from a small crystallization/melting peak that can be seen only in the 4:1 trace at $20/41^\circ\text{C}$. This weak feature associates with the crystallization/melting of confined PEO.³¹ Among all compositions, the 4:1 has the highest melting temperature and the highest heat of fusion for the complex. However, the 3:1 compound is the first where PEO crystallization is completely suppressed leaving the complex as the only ordered (crystalline) structure.

For the 22 kg/mol $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolyte (Figure 3), the cooling trace of the 12:1 compound shows a

broad exothermic peak around 22°C and a glass temperature at -49°C . On subsequent heating, the glass temperature shifts to -43°C and is followed by an exothermic peak at 27°C signifying cold crystallization of PEO/complex formation followed by a PEO melting endotherm at 57°C . Just above the PEO melting temperature there exist a broad and asymmetric endotherm with a peak at 120°C signifying the melting of the complex. As with the lower molecular weight sample, the cooling and heating traces of the 8:1 compound are qualitatively similar, but the complex melts at a higher temperature (126°C). The cooling trace of the 6:1 compound exhibits a broad exotherm with peaks at 88 and 76°C due to complex formation and a PEO crystallization peak at 24°C . The glass temperature is now less evident in this representation because the exothermic peaks are very strong. On heating, PEO melts at 46°C followed by a broad complex melting with a peak at 140°C . The traces of the compounds with 4:1, 3:1, and 2:1 ratios are qualitatively similar in that they reveal that PEO is unable to crystallize (a small crystallization/melting peak may be seen in the 4:1 trace at $24/42^\circ\text{C}$). As with the lower molecular weight PEO, the 4:1 composition is the one with the highest melting temperature, and this despite the small fraction of crystallizable PEO. On the other hand, the 3:1 composition is the first where the complex is the only ordered phase; however, its melting temperature is reduced as compared to the 4:1. Results for the higher molecular weight (Figure S1) revealed the absence of PEO cold crystallization and a better-defined (i.e., narrower) complex melting peak at the 4:1 composition.

These results (including results from the higher molecular weight (Figure S1)) are compiled in Figure 4. The figure

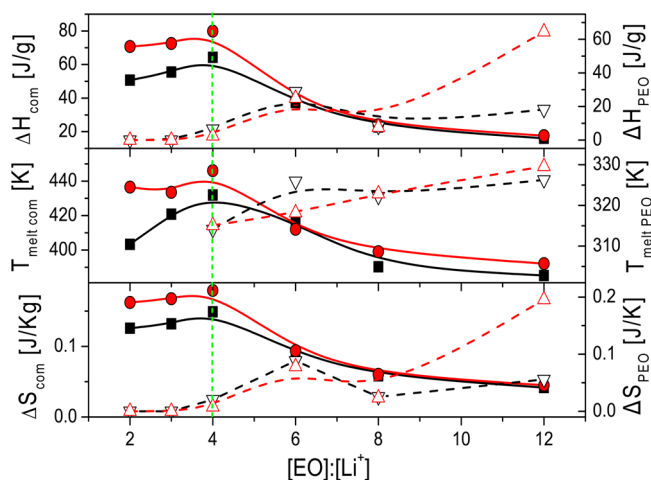


Figure 4. Change in enthalpy (top), melting temperature (middle), and change in entropy (bottom) for the melting point of the complex (left Y-axis) and of PEO crystals (right Y-axis) in $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ electrolytes. The black symbols correspond to polymer electrolytes with PEO $M_w = 14$ kg/mol, while red symbols to $M_w = 22$ kg/mol. Lines are guides to the eye.

depicts the melting temperatures of the complex (i.e., the *liquidus*) and of the PEO (*solidus*), the associated heats of fusion, and the calculated change in entropy at the transition (as $\Delta S = \Delta H/T$). The *stoichiometric* composition is the one that “consumes” all reagents, and this corresponds to the 3:1 ratio. At this composition PEO is unable to crystallize. On the other hand, if we define the *stoichiometric* composition as the

one with the most stable complex (i.e., with the highest heat of fusion), then this can be identified as the 4:1, independent of the molecular weight. The calculated change of entropy at the transition is also the highest for the 4:1 ratio. Assuming similar entropy for the melt state for the different compounds we conclude that the 4:1 compound forms the most ordered complex. Notice that the latter method was used in constructing the existing phase diagram of $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolyte (Figure 1),⁹ but the results were based on the first rather than the second heating runs (this explains the differences with the data of Figure 4). Earlier diffraction studies,¹⁰ however, provided an upper limit to the $[\text{EO}]:[\text{Li}]$ ratio at the stoichiometric composition as 3.13. We therefore employ this definition $(\text{PEO})_3\text{LiCF}_3\text{SO}_3$ for the stoichiometric complex.

Crystal Structure. The crystal structure of the $(\text{PEO})_3\text{LiCF}_3\text{SO}_3$ polymer electrolyte is well-known.¹⁰ PEO adopts a helical conformation parallel to the crystallographic b -axis, and Li^+ ions are encapsulated within the PEO chain. Lithium is coordinated by five oxygens, composed of three PEO oxygens and one oxygen from each of two adjacent CF_3SO_3^- groups. Notably, lithium coordination is solely intramolecular, meaning that there are no lithium bridges between different chains. We mention this difference from other polymer electrolytes, such as $(\text{PEO})_x\text{LiXF}_6$ ($X = \text{P}, \text{As}, \text{Sb}$).^{32,33} Figure 5 gives the WAXS spectra of the

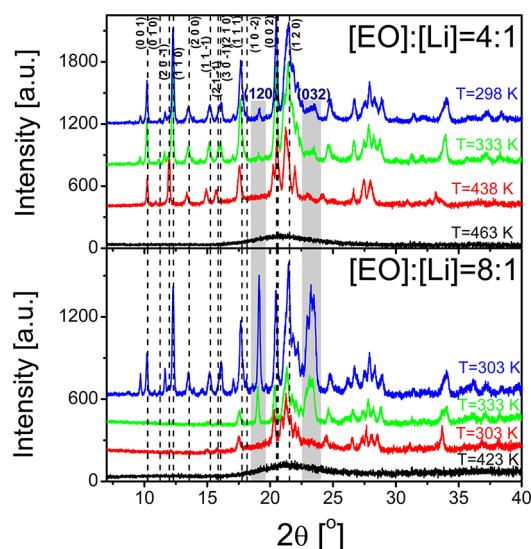


Figure 5. (top) WAXS spectra of $(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ ($M_w = 14$ kg/mol) polymer electrolyte at different temperatures as indicated. (bottom) WAXS spectra of the $(\text{PEO})_8\text{LiCF}_3\text{SO}_3$ ($M_w = 14$ kg/mol) polymer electrolyte at different temperatures as follows: 303 K (as prepared, blue line), 333 K (obtained by heating, green line), 423 K (obtained on heating, black line), and 303 K (obtained on cooling from 423 K, red line). The gray areas give the positions of the most intense reflections from the monoclinic unit cell of PEO. The dashed lines give the positions and (hkl) indices of the unit cell of the crystalline complex.

$(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ polymer electrolyte ($M_w = 14$ kg/mol) at different temperatures. The diffraction peaks are consistent with the known unit cell of $(\text{PEO})_3\text{LiCF}_3\text{SO}_3$ (monoclinic unit cell with $a = 1.677$ nm, $b = 0.861$ nm, $c = 1.007$ nm, and $\beta = 121.02^\circ$), suggesting that the complex in the nonstoichiometric compositions has exactly the same crystal structure.¹⁰ A small

fraction of PEO remains uncomplexed as seen in the DSC trace in Figure 2, and this crystallizes in the usual monoclinic unit cell of PEO as evidenced by the (120) and (032) reflections.³⁴ On heating to 333 K, the crystal structure of the complex remains intact. Only by further heating to 438 K some of the peaks lose intensity, reflecting partial melting of the complex at this temperature (Figure 2). Further heating to 463 K results in the complete melting of the structure and to a liquid-like structure factor, confirming the amorphous systems above the liquidus temperature.

Figure 5 provides also with the WAXS spectra of the $(\text{PEO})_8\text{LiCF}_3\text{SO}_3$ polymer electrolyte at different temperatures. The crystal structure of the complex is identical to the 4:1 and 3:1 compositions, suggesting the same crystal structure that is in accord with the known phase diagram (Figure 1). At this ratio, PEO crystallizes as seen by the intense (120) and (032) reflections of PEO in agreement with DSC (Figure 2). However, heating to 333 K brings several structural changes. The intensity of the reflections from the PEO crystals and the complex are now weak, and this is in agreement with the continuous melting of the complex starting already in the vicinity of PEO melting (solidus) as seen in DSC (Figure 2). Further heating to 423 K (above the liquidus temperature) followed by cooling and measurement at 303 K results in a diffraction pattern that is not identical to the one mentioned on heating, suggesting slow crystallization kinetics of PEO and of the complex.

In conclusion, the crystal structure of the complex is identical to the $(\text{PEO})_3\text{LiCF}_3\text{SO}_3$ polymer electrolyte independent of the $[\text{EO}]:[\text{Li}]$ content. Heating the nonstoichiometric mixtures results in progressive melting of the complex and the incorporation of ions in the amorphous phase whereas the complex formed at or near the stoichiometric composition remains stable up to the liquidus temperature. These structural features are expected to influence the ionic conductivity.

DC Conductivity. The dc conductivities of the $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolytes with molecular weights of 14 and 22 kg/mol are depicted in Figures 6 and 7, respectively, as a function of temperature. Figure 6 shows that the substoichiometric $(\text{PEO})_2\text{LiCF}_3\text{SO}_3$ polymer electrolyte, i.e., the one with the highest lithium ion content, has the lowest conductivity at all temperatures investigated. This reflects the restricted ion motion by the solid phases at the peritectic point (Figure 1).⁹ On the other hand, $(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ with a composition near the stoichiometric one has an appreciable conductivity: from 3×10^{-7} S/cm at ambient temperature to 5×10^{-5} S/cm just before melting of the stable complex. This is remarkable, as at this composition nearly all PEO (Figures 2 and 4) is complexed to Li^+ and CF_3SO_3^- . It is worth mentioning that such conductivities (i.e., with a lower threshold of 10^{-7} S/cm) are required in some devices (electrochromic displays, smart windows), whereas much higher values are required for lithium batteries. This clearly shows that ionic conduction can take place within the intramolecularly coordinated Li^+ and CF_3SO_3^- ions. This situation in the $(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ polymer electrolyte has similarities to the high conductivity found in $\text{PEO}_6\text{LiSbF}_6$ polymer electrolyte with the stoichiometric composition. However, there^{32,33} ion transport was facilitated through cylindrical channels formed by pairs of PEO chains, in contrast to the present system where ions are complexed intramolecularly. The role of local polymer mobility on the dynamics of intramolecularly complexed ions will be discussed below.

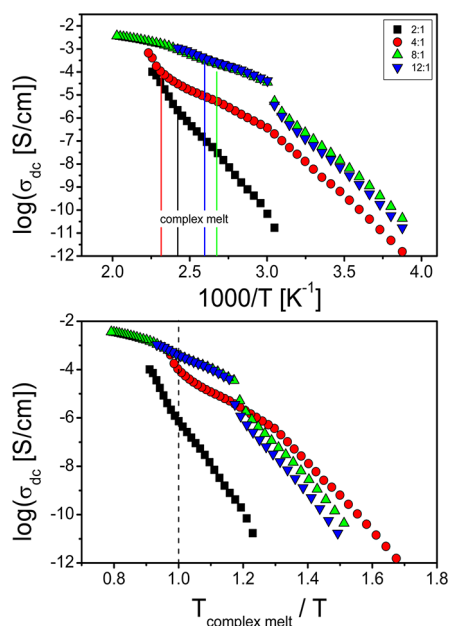


Figure 6. (top) Dc conductivities of the polymer electrolytes $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ ($M_w = 14$ kg/mol) with $x = 2$ (squares), 4 (spheres), 8 (up triangles), and 12 (down triangles). Vertical lines give the melting temperatures (liquidus) of the respective complex. (bottom) The same data in a temperature-scaled representation with respect to the melting point of the complex.

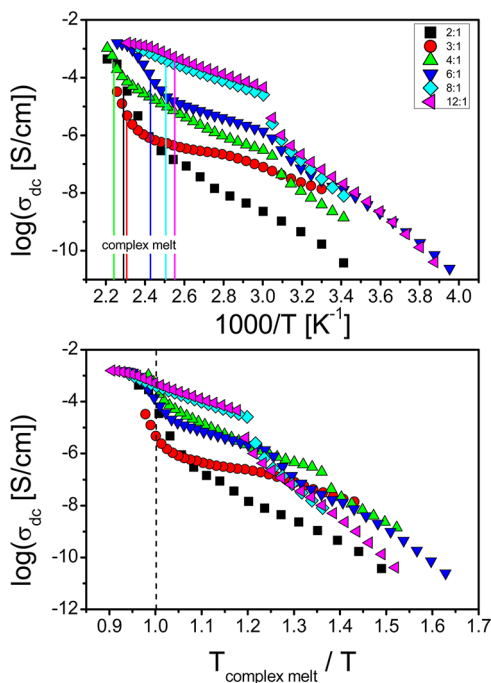


Figure 7. (top) Dc conductivities of the polymer electrolytes $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol) with $x = 2$ (squares), 3 (spheres), 4 (up triangles), 6 (down triangles), 8 (rhombi), and 12 (left triangles). Vertical lines give the melting temperatures of the respective complex. (bottom) The same data in a temperature-scaled representation with respect to the melting point of the complex.

Some insight into the type of motions that give rise to ionic mobility will be provided later with respect to Figure 13. Ionic conduction is thermally activated but does not follow a single Vogel–Fulcher–Tammann or Arrhenius dependence. This is

the case at all compositions investigated and contrasts with some earlier reports^{11,19,21,22} over a narrower T -range claiming either a VFT or an Arrhenius dependence.

For the more dilute polymer electrolytes with compositions 8:1 and 12:1 one can notice an appreciable (by 2 orders of magnitude) increase of the dc conductivity at the PEO melting temperature. In addition, $\sigma_{dc}(T)$ has a weaker temperature dependence above the PEO melting temperature than below and shows a second smaller step at the liquidus temperature. Since the highest conductivity is obtained above the liquidus temperature, an alternative representation of the conductivity data is by using reduced temperatures relative to the liquidus. This is also depicted in Figure 6 and reveals that the 12:1 composition has a higher dc conductivity. The conductivities for the $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolytes with the higher molecular weight (22 kg/mol) (Figure 7) display similar features. The polymer electrolyte with the stoichiometric $(\text{PEO})_3\text{LiCF}_3\text{SO}_3$ composition possesses appreciable conductivities. Again, the highest conductivities are associated with the 12:1 composition.

The conductivities for the different molecular weights and the different compositions are summarized in Figure 8. The

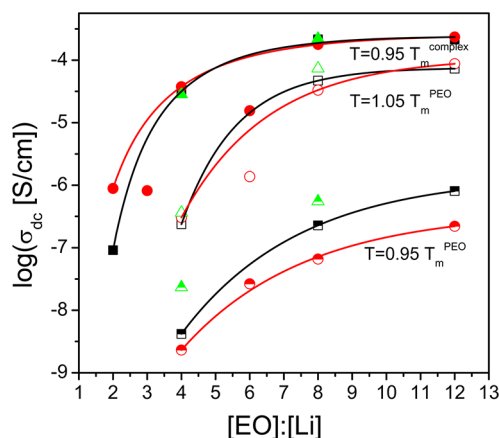


Figure 8. Dc conductivity of the polymer electrolytes $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ as a function of ether oxygen to Li cation ratio plotted at different reduced temperatures: $T = 0.95T_m^{\text{complex}}$, i.e., just below the melting of complex (filled symbols), $T = 1.05T_m^{\text{PEO}}$, i.e., just above the PEO melting (open symbols), and $T = 0.95T_m^{\text{PEO}}$, i.e., just below PEO melting (half-filled symbols). The different symbols correspond to the different molecular weights as follows: (squares) $M_w = 14$ kg/mol, (spheres) $M_w = 22$ kg/mol, and (triangles) $M_w = 50$ kg/mol.

results are discussed at three reduced temperatures: just below the PEO melting temperature ($T = 0.95T_m^{\text{PEO}}$), just above the PEO melting ($T = 1.05T_m^{\text{PEO}}$), and below the melting of the complex ($T = 0.95T_m^{\text{complex}}$). Remarkably, it is the 12:1 composition that has the highest dc conductivity at all temperatures. This suggests the importance of the amorphous phase where ions are continuously incorporated following progressive melting of the stoichiometric complex. In this view, the complex provides an ion sink for charge transport.

The molecular weight dependence of conductivity is discussed in Figure 9 where the data for the $(\text{PEO})_8\text{LiCF}_3\text{SO}_3$ polymer electrolyte are plotted as a function of temperature. There is not any significant molecular weight dependence except perhaps below the PEO melting, where the PEO molecular weight enters the nucleation and growth kinetics of

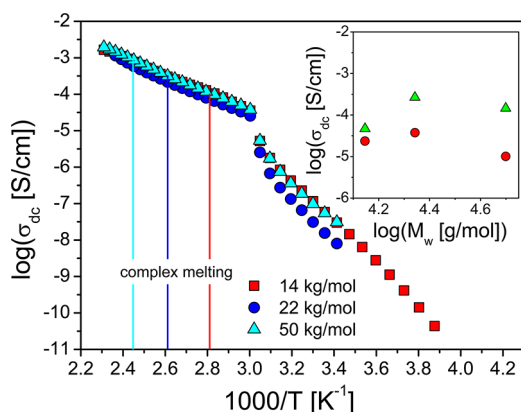


Figure 9. Dc conductivity of $(\text{PEO})_8\text{LiCF}_3\text{SO}_3$ plotted for different molecular weights of PEO. Vertical lines give the melting temperatures of the respective complex. Inset: dc conductivities for $(\text{PEO})_8\text{LiCF}_3\text{SO}_3$ (up triangles) and $(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ (red spheres), at the same reduced temperature, $T = 0.95T_{\text{complex}}$, as a function of PEO molecular weight.

PEO crystals. In the inset we compare the dc conductivities of the different molecular weights at the same reduced temperature of $0.95T_{\text{m}^{\text{complex}}}$. The molecular weight dependence is very weak, implying that chain length is not an essential parameter for ion transport for sufficiently high molecular weights, in agreement with earlier reports.^{35,36}

In summary, the T dependence of dc conductivity is neither Arrhenius nor VFT; it is much more complex and reflects the continuous underlying structural changes. Although the main contribution to ion motion is through the amorphous phase that is enriched with LiTf ions, there is some additional contribution coming from the stoichiometric complex itself. The $(\text{PEO})_{12}\text{LiCF}_3\text{SO}_3$ polymer electrolyte with the higher crystallinity is the one with the highest conductivity at all temperatures and compositions investigated. This paradoxical situation has been recognized earlier.⁶ It suggests that facile ion transport requires (i) continuous supply of ions from the ion sink (continuous melting of complex) and (ii) melted PEO domains of low viscosity where the ions are incorporated and transported.

Viscoelastic Behavior. The viscoelastic properties of two $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolytes with $x = 12$ and 4 have been examined as being representative of the other compositions. First, the linear viscoelastic range was identified by performing strain amplitude sweeps at selected temperatures. We found that the complex is very susceptible to strain, and soon the behavior turns nonlinear (Figure S5). In the remaining measurements we have employed low strain amplitudes corresponding to the linear viscoelastic regime. Figure 10 gives the storage and loss moduli for the two compositions under isochronal ($\omega = 10$ rad/s) conditions. The moduli of the 12:1 composition display an elastic response at temperatures below 50 °C that is followed by a regime where $G' \sim G''$ over a broad temperature range, followed by a transition to the liquid state. The (elastic) properties of the low temperature phase are determined by both the crystalline PEO and the stoichiometric complex. The increased glass temperature relative to bulk PEO (DSC) suggests also the presence of a “amorphous” but coordinated PEO phase. Upon PEO melting, the nearly parallel dependence of the moduli reflect the continuous melting of the complex in agreement with the DSC (Figure 3) and WAXS (Figure 5) results. This should be

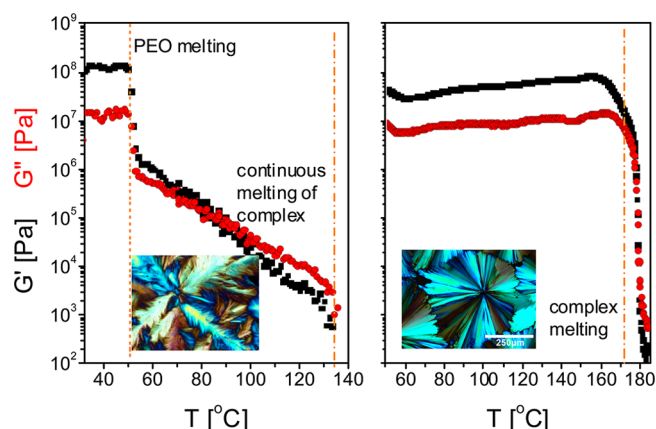


Figure 10. Storage (squares) and loss (spheres) moduli of $(\text{PEO})_{12}\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol) on the left and $(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol) on the right, plotted as a function of temperature. Data were obtained at $\omega = 10$ rad/s with a heating rate of 2 K/min. The dashed and dash-dotted vertical lines indicate respectively the melting of PEO crystals and the melting of the complex. The insets give POM images of the superstructure formation involving the crystalline complex. The scale bar indicates 250 μm .

contrasted with the 4:1 composition where PEO is largely unable to crystallize, leaving the complex as the single ordered phase that remains stable up to the liquidus temperature. The complex is also characterized by a predominantly elastic response up to the melting temperature. Interestingly, space-filling “spiky” vs “multisheave” spherulitic superstructures are seen in the POM images of the 4:1 and 12:1 compositions.³⁷ Both superstructures associate with the growth of the crystalline complex at stoichiometric and more dilute compositions.

Detailed information on the viscoelastic properties can be obtained by studying the frequency response of the different phases. Figure 11 gives the frequency dependence of the storage and loss moduli for the 12:1 and 4:1 compositions at some selected temperatures. For the 12:1 composition, the frequency response at 40 °C is purely elastic with $G' \gg G''$ as reflecting the response of the crystalline PEO and of the complex. At 75 °C, $G' > G'' \sim \omega^{1/3}$ whereas at 100 °C, $G' \sim G'' \sim \omega^{1/3}$. This results from the appearance of a new ultraslow relaxation process related to a morphological rearrangement (complex). However, a complete relaxation of the complex cannot be observed within the accessible frequency and temperature window. This is reminiscent to the response of (i) evolving systems at the gel point and to (ii) nanophase separated block copolymers. In the former,^{38,39} the initial sol state is characterized by $G'(\omega) \sim \omega^2$, $G''(\omega) \sim \omega^1$ and the final gel spectrum by $G'(\omega) \sim G''(\omega) \sim \omega^0$, whereas at the gel point the spectrum shows a power law dependence. In block copolymers,^{40–42} depending on the nanophase morphology similar dependencies have been observed ($G'(\omega) \sim G''(\omega) \sim \omega^{1/2}$ for a lamellar mesophase and $G'(\omega) \sim G''(\omega) \sim \omega^{1/4}$ for a cylindrical mesophase). At 130 °C, i.e., above the liquidus temperature, a typical terminal relaxation is observed ($G'(\omega) \sim \omega^2$, $G''(\omega) \sim \omega^1$). For the 4:1 composition, the response at 100 °C is predominantly elastic with some indication for a slow relaxation associated with the reorganization of the complex. Thus, the mechanical properties of the complex at near stoichiometric composition are fundamentally different from the structure formed in the more dilute polymer electrolytes

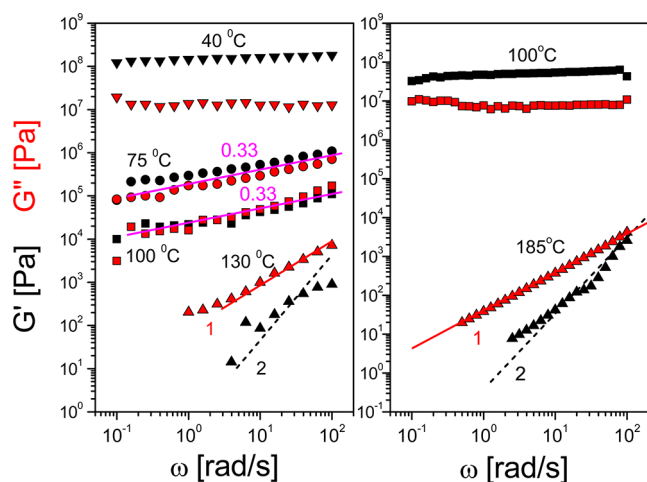


Figure 11. (left) Frequency dependence of the storage (black symbols) and loss (red symbols) moduli, of $(\text{PEO})_{12}\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol), at different temperatures: $T = 40$ °C (triangles) probing the crystalline PEO, crystalline complex, and amorphous phase with a solidlike response; $T = 75$ and 100 °C probing the viscoelastic properties of the complex and amorphous phase with $G' \sim G'' \sim \omega^{1/3}$ and at $T = 130$ °C, probing the terminal relaxation. (right) Frequency dependence of the storage (black symbols) and loss (red symbols) moduli of $(\text{PEO})_4\text{LiCF}_3\text{SO}_3$ ($M_w = 22$ kg/mol), at different temperatures: $T = 100$ °C (squares) probing the complex with an elastic response and at $T = 185$ °C probing the terminal relaxation. Lines with slopes 1 and 2 are drawn which indicate the range used to estimate the terminal relaxation times.

(predominantly elastic vs viscoelastic), and this despite the same crystal structure of the complex at all compositions (Figure 4). Clearly, the presence of the amorphous phase in the nonstoichiometric compositions influences the viscoelastic properties. The present results are in contrast with earlier identifications of the complex as “elastomeric”.¹¹

Summarizing the linear viscoelastic properties, we find that the different phases have distinct viscoelastic signatures. At or near stoichiometric composition the complex has a predominantly elastic response with some indication for an ultraslow relaxation, whereas the structure formed at more dilute concentrations (crystalline complex and amorphous ion-rich phase) has a viscoelastic response.

Ion Mobility and Local Segmental Dynamics. There is consensus that ion mobility in SPEs is coupled to the polymer segmental motion. To explore this in more detail, we studied the local dynamics with DS in comparison to the DSC glass temperatures. In DS we found that the polymer segmental dynamics is governed by the ionic motion. Some representative ϵ^* , M^* , and σ^* data representations are shown in Figure S8. In bulk PEO, there is a step in the dielectric permittivity at higher frequencies that reflects the polymer segmental dynamics and a second step at lower frequencies that corresponds to the crossing of the real and imaginary parts which is associated with the conductivity contribution (due to impurities).²⁷ Such two separate steps in the permittivity spectra—and the corresponding processes in the dielectric loss curves—were never observed in the $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolytes. Instead, a single step in ϵ' that takes place at the vicinity of crossing with the ϵ'' , suggesting the complete coupling of the “segmental” relaxation to the ion mobility (Figure 12). This is better shown in Figure S9, where the analysis of the segmental relaxation is based on the derivative of ϵ' . The data refer to the polymer electrolyte

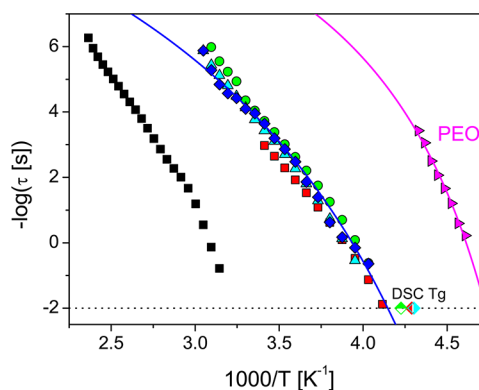


Figure 12. Segmental relaxation times of bulk PEO (magenta right-triangles) and of $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ ($M_w = 14$ kg/mol) with $x = 2$ (black squares), $x = 4$ (red squares), $x = 8$ (green circles), and $x = 12$ (cyan triangles). The rhombi at $\tau = 100$ s represent the glass temperature as obtained from DSC (symbols have the corresponding color code).

$(\text{PEO})_{12}\text{LiCF}_3\text{SO}_3$ and show the coupling of segmental to the ionic motion. Furthermore, the α -process of bulk PEO is now shifted to higher temperatures, and this is in agreement with the DSC results on the liquid-to-glass temperature. This suggests a different nature for the “segmental” relaxation in the polymer electrolytes as compared to the homopolymer PEO. For the 4:1 composition, i.e., near the stoichiometric composition, nearly all ions are complexed with the polymer chains. This effectively stiffens the polymer backbone and gives rise to the increased glass temperature T_g as observed in DSC. Since any local segmental motion would have to displace the intramolecularly coordinated cations and anions, it is not surprising that the modified backbone dynamics is coupled to the ion motion. What is surprising, however, is that the same local dynamics are observed in the nonstoichiometric ratios of 8:1 and 12:1. We speculate that the presence of PEO crystals effectively enrich the ion concentration in the amorphous phase. Indeed, a PEO crystallinity of 33% in the 12:1 compound ($M_w = 22$ kg/mol) results in an effective $[\text{EO}]:[\text{Li}]$ composition of 8:1 in the amorphous phase. Similarly, for the 6:1 composition with 14% crystalline PEO, the effective composition outside the crystalline PEO is 5:1.

With the exception of the 2:1 polymer electrolyte where a different complex is formed,⁹ the modified segmental dynamics in all $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ are nearly identical. The new process conforms to the Vogel–Fulcher–Tammann (VFT) equation:

$$\tau = \tau_0 \exp\left(\frac{B}{T - T_0}\right) \quad (3)$$

where τ_0 ($= 10^{-12}$ s) is the relaxation time in the limit of very high temperatures, $B = 2520$ K is the activation parameter, and $T_0 = 163$ K is the “ideal” glass temperature. This should be compared with the segmental process of bulk PEO³⁸ with $\tau_0 = 10^{-12}$ s, $B = 1030$ K, and $T_0 = 179$ K.

More insight into the characteristic length scales of ion motion can be extracted by combining the effective times of ion motion with literature diffusion data for the anion and cation. Pulse field-gradient (PFG) NMR cation and anion diffusion coefficients in the $(\text{PEO})_8\text{LiCF}_3\text{SO}_3$ polymer electrolyte were

reported¹² in the temperature range from 423 to 441 K, i.e., at temperatures below and above the liquidus temperature. We have employed the reported Arrhenius T dependencies for the anion and cation diffusion coefficients and the single characteristic time of ion motion extracted from DS (Figure 12). We further assume the validity of the Einstein–Smoluchowski equation, $D = \langle r^2 \rangle / 6\tau$, relating the microscopic details of particle diffusion (the mean-squared displacement, $\langle r^2 \rangle$, and characteristic time of motion, τ) to the macroscopic properties (such as the diffusion coefficient, D) valid for the Brownian motion of suspended particles. Certainly, ionic diffusion below the liquidus temperature is very complex comprising both ion motion complexed with the polymer chains and ion transport in amorphous domains. The former motion can hardly be considered as homogeneous diffusion in violation of a basic assumption in the SE equation. Nevertheless, since the diffusion within the amorphous domains is expected to have a much greater contribution (Figures 6 and 7), we employ this equation to obtain a rough estimate of the ion length scale, i.e., the mean free path, $\langle r^2 \rangle^{1/2}$, which is discussed in Figure 13. The

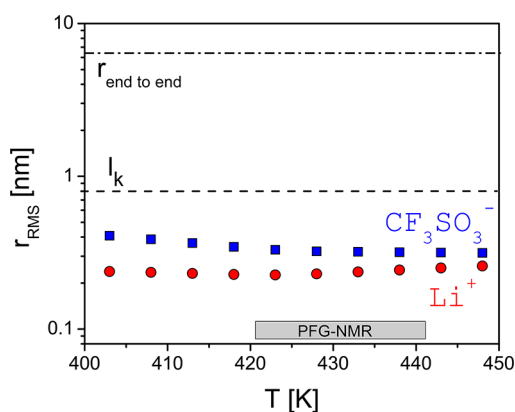


Figure 13. Mean free path of ion motion as a function of temperature for $(\text{PEO})_8\text{LiCF}_3\text{SO}_3$ ($M_w = 14$ kg/mol) obtained within the assumptions of the Einstein–Smoluchowski equation: (blue squares) CF_3SO_3^- , (red circles) Li^+ . The gray scale bar indicates the range of NMR measurements.¹² The experimental symbols are calculated based on the validity of the above equation and extrapolations of the NMR diffusion coefficients to lower and higher temperatures (see text). The dashed and dash-dotted lines give the Kuhn length (l_k) and polymer end-to-end distance (r), respectively.

figure depicts the range of the actual NMR measurements and reveals a higher mean free path for the anion as compared to the cation (the latter reflects the higher transference number for the cation). But interestingly, it shows that both length scales are of subsegmental scale, i.e., below the polymer Kuhn length⁴³ (the Kuhn length in the present case is expected to be even larger). Within the above-mentioned assumptions, this result may reflect local ion jumps of the size of the ionic radii, as the underlying process giving rise to the measured conductivity. This point, however, requires further investigations by combining PFG-NMR with DS in a number of compositions and for different SPEs.

IV. CONCLUSION

The current investigation relates the structural features of the archetypal polymer electrolyte $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ to the local ion and polymer relaxation and to the overall ionic conductivity. The crystal structure of the complex is identical

to the $(\text{PEO})_3\text{LiCF}_3\text{SO}_3$ polymer electrolyte independent of the $[\text{EO}]:[\text{Li}]$ content. Heating the nonstoichiometric compositions results in progressive melting of the complex. This is a key structural feature as it results in the incorporation of ions in the amorphous phase. On the other hand, the complex formed at or near the stoichiometric composition remains stable up to the liquidus temperature.

We found two mechanisms of ionic conduction. The first one is through the intramolecularly coordinated Li^+ and CF_3SO_3^- ions within the crystalline complex with a maximum conductivity of 5×10^{-5} S/cm just before the liquidus temperature. This is the sole mechanism of ion conduction in the stoichiometric or near-stoichiometric complex. The second—and more effective mechanism—is through the amorphous phase where ions are continuously incorporated following progressive melting of the stoichiometric complex. In the second case, the complex provides an ion sink for charge transport. Among different compositions, the $(\text{PEO})_{12}\text{LiCF}_3\text{SO}_3$ polymer electrolyte with the highest crystallinity is the one with the highest conductivity at all temperatures and compositions investigated. This seemingly paradoxical situation suggests that facile ion transport requires the continuous supply of ions from the ion sink (the complex) and melted PEO domains at low viscosity where ions are incorporated and transported. As a result of the continuous structural changes the temperature dependence of dc conductivity cannot be represented by an Arrhenius or a VFT law as experimentally observed. Local polymer relaxation and ionic mobility were found to be completely coupled at all compositions. By combining the present DS results with earlier NMR diffusion coefficients for the cation and the anion, it suggests that local ion jumps at subsegmental level are responsible for the measured conductivity.

The linear viscoelastic properties of the polymer electrolytes revealed that the different phases have distinct viscoelastic signatures. The complex formed at or near-stoichiometric composition has a predominantly *elastic* response whereas the more dilute compositions (consisting of the crystalline complex and an ion-containing amorphous phase) have a *viscoelastic* response and an ultraslow relaxation. It further points out to the viscosity needed for better ion transport.

The results presented here provide the necessary background material in discussing ion transport in other SPEs composed of more heterogeneous systems, like block copolymers. In these systems ion transport takes place selectively through the PEO nanophase; thus, a comparison with the $(\text{PEO})_x\text{LiCF}_3\text{SO}_3$ polymer electrolyte is required. We will return to this point in the near future.

■ ASSOCIATED CONTENT

Supporting Information

Details on DSC, TGA, and DS measurements and analyses. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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